SCH (PCBA) Part Number: **DT10965**

SCHEMATIC MAJOR REVISION HISTORY

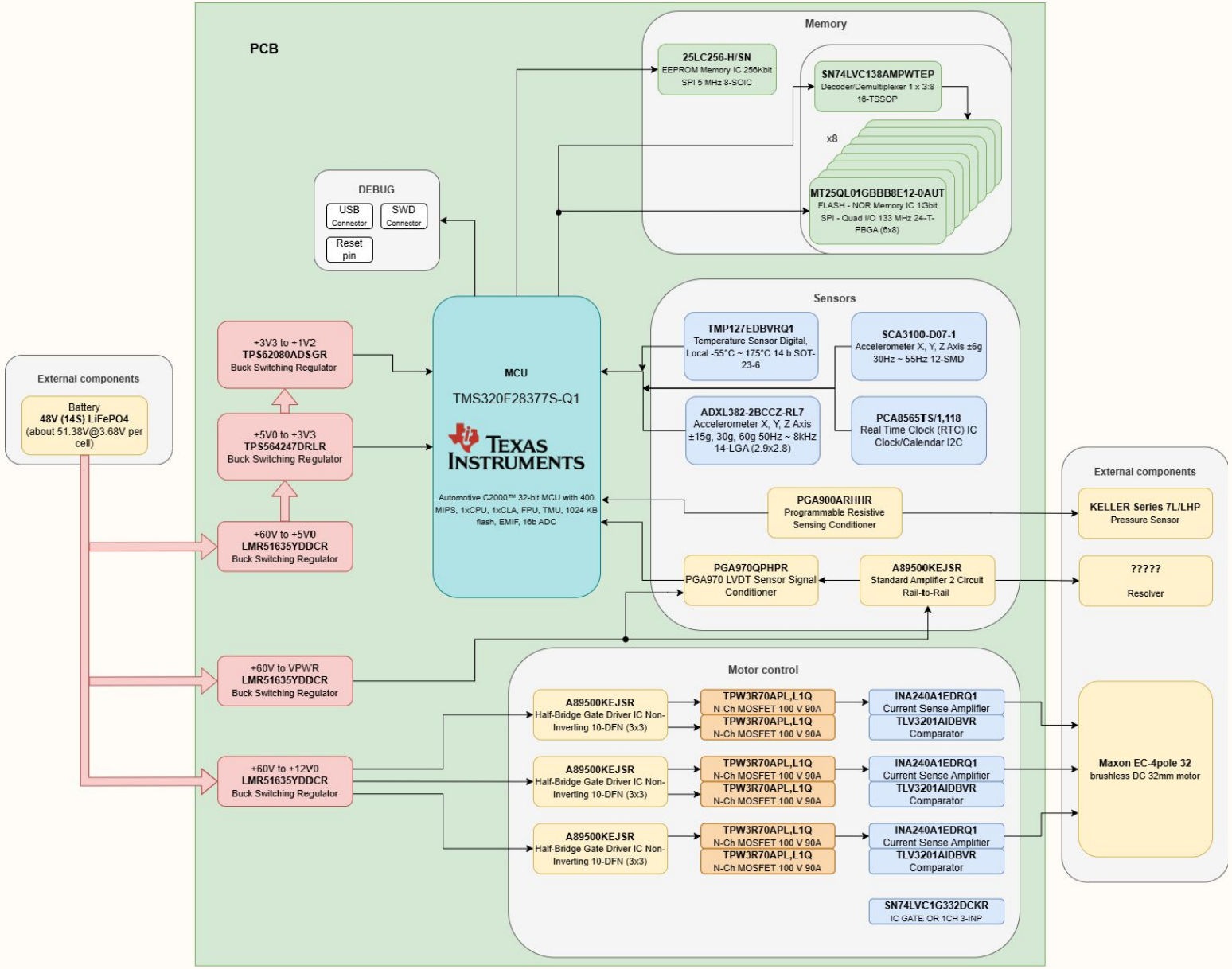
REVISION	SUBREVISION	CHANGE DESCRIPTION	YYYY/MM/DD

PCB Part Number: **DT10966**

PCB MAJOR REVISION HISTORY

REVISION	SUBREVISION	CHANGE DESCRIPTION	YYYY/MM/DD

Title *MOTORCONTROLLER*			*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:44	Sheet 1 of 16	

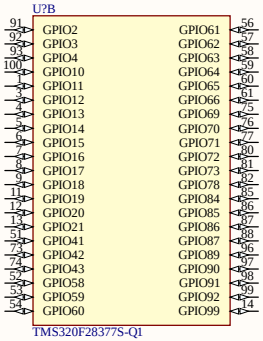


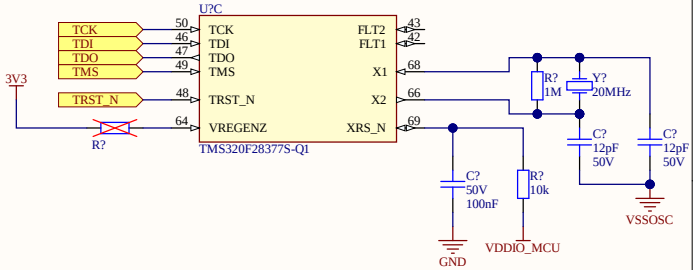
Title	**MOTORCONTROLLER**		Droid Technologies
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:44	Sheet 2 of 16	

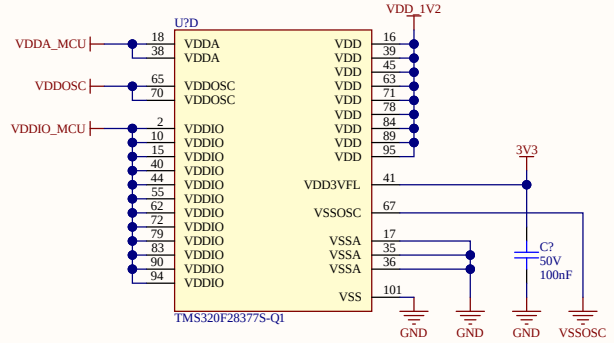
MCU

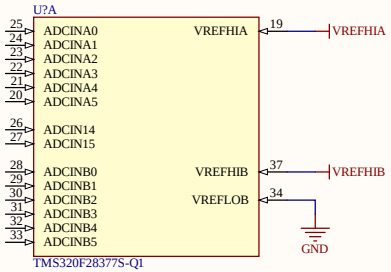
3V3
VDD_1V2
VREFHIA
VREFHIB

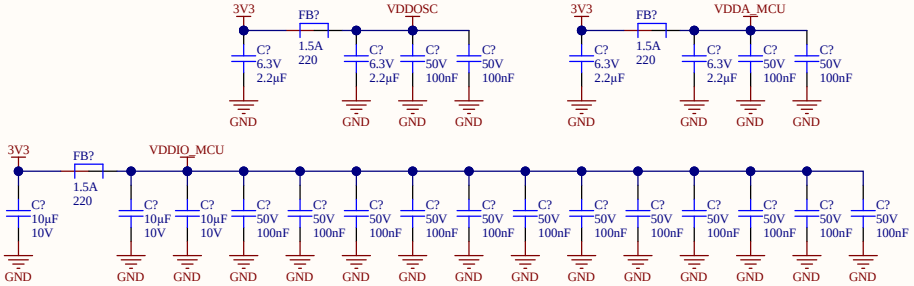
Device supply voltage, I/O, VDDIO = +3.14V to +3.47V (30mA(max))
Device supply voltage, VDD = +1.14V to +1.26V (400mA(max))
Analog supply voltage, VDDA = +3.14V to +3.47V (20mA(max))
IDD3VFL = 40mA(max) @
Ambient operating temperature (max) = –40°C to 125°C free-air

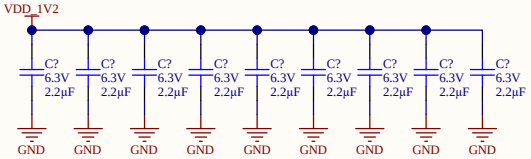




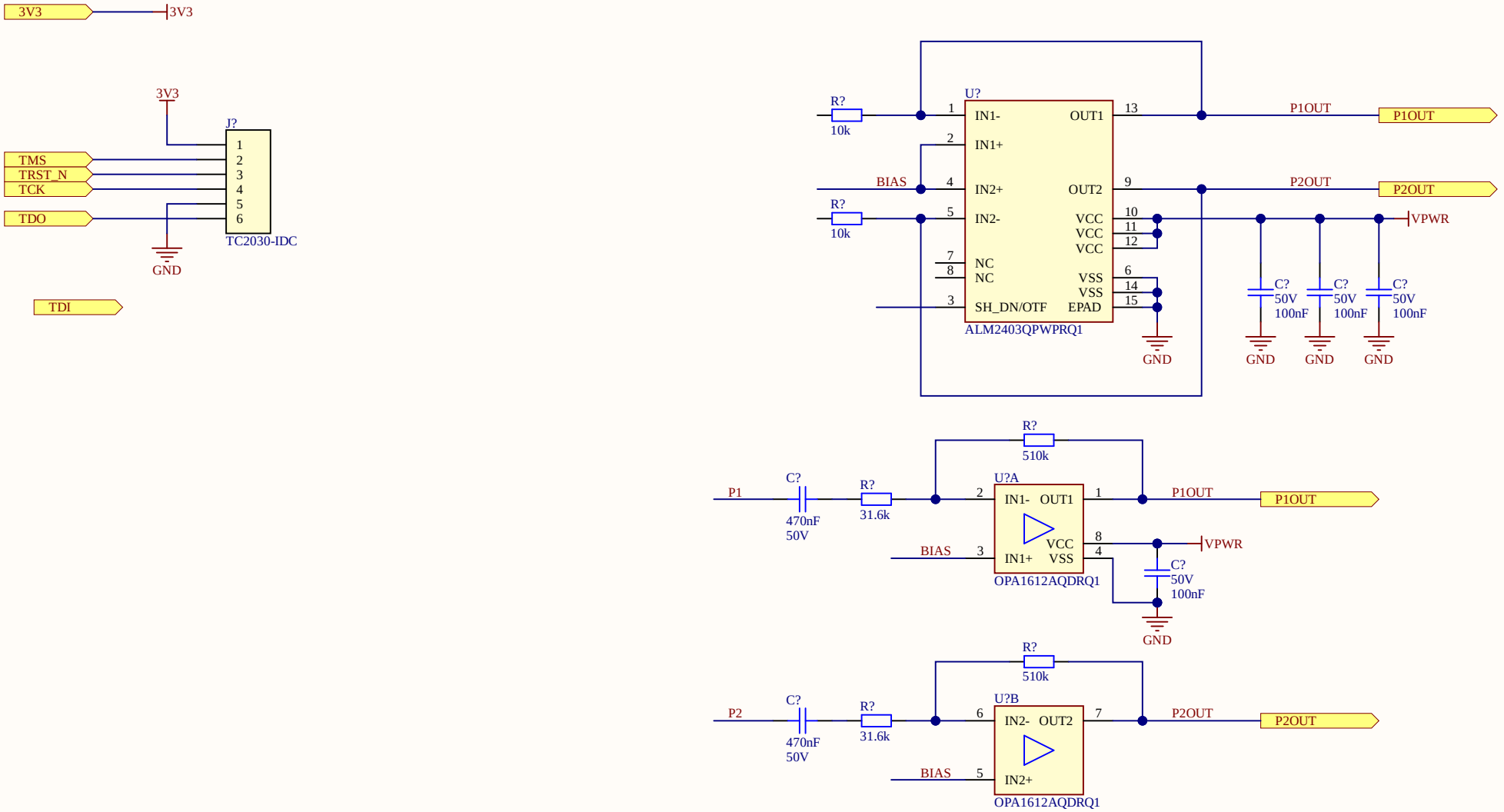






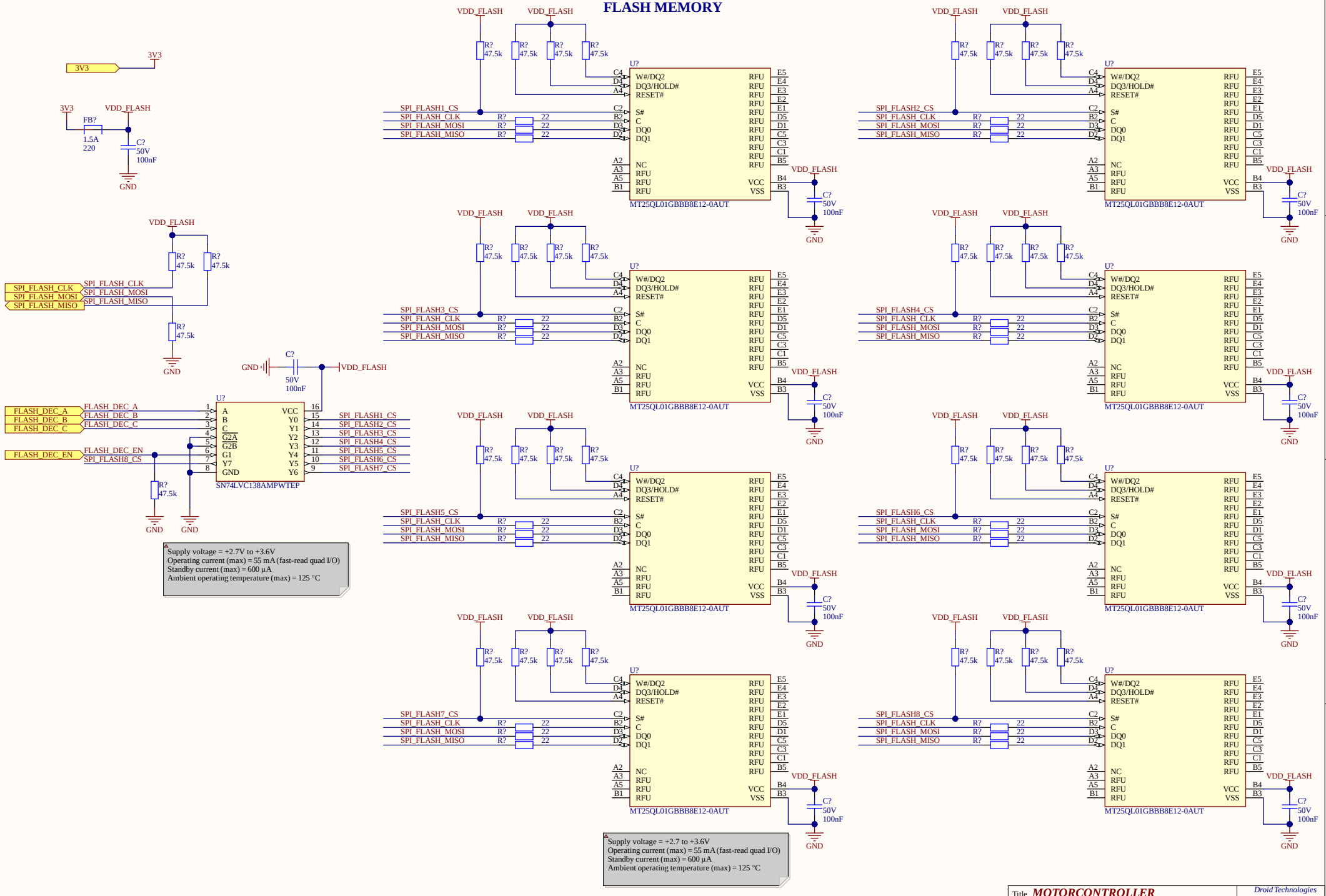


Title: **MOTORCONTROLLER**			Droid Technologies
Size: A3	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:45	Sheet 3 of 16	



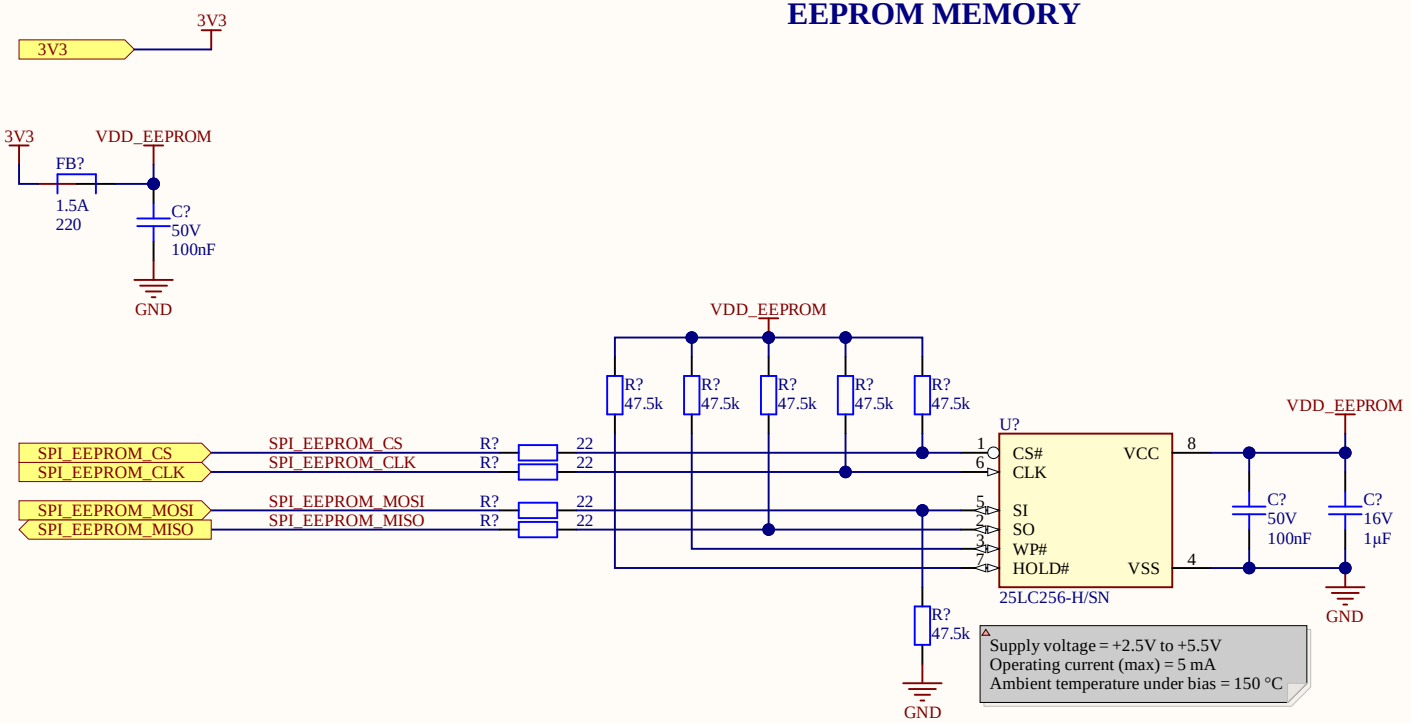
Title	***MOTORCONTROLLER***		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:45	Sheet 4 of 16	

FLASH MEMORY



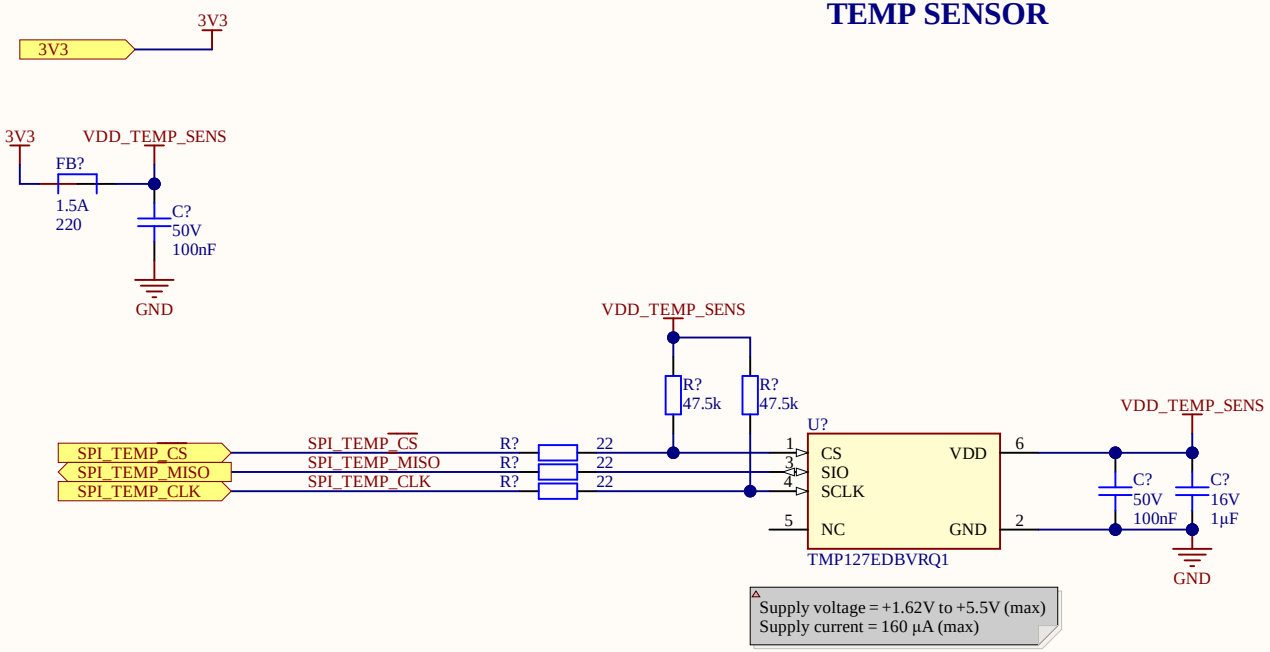
Title	MOTORCONTROLLER		Droid Technologies
Size: A3	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:45	Sheet 5 of 16	

EEPROM MEMORY



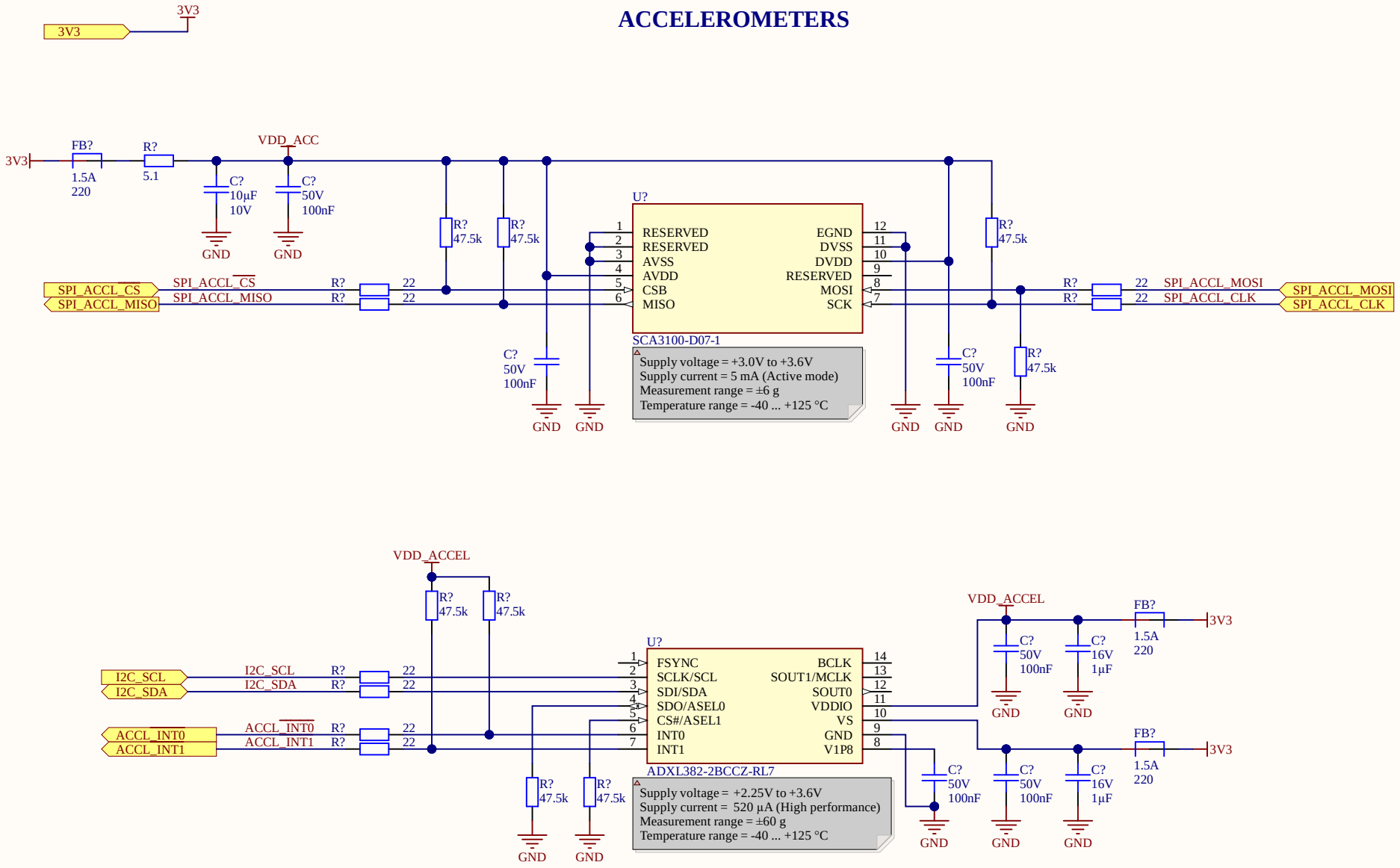
Title	***MOTORCONTROLLER***		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:45	Sheet 6 of 16	

TEMP SENSOR

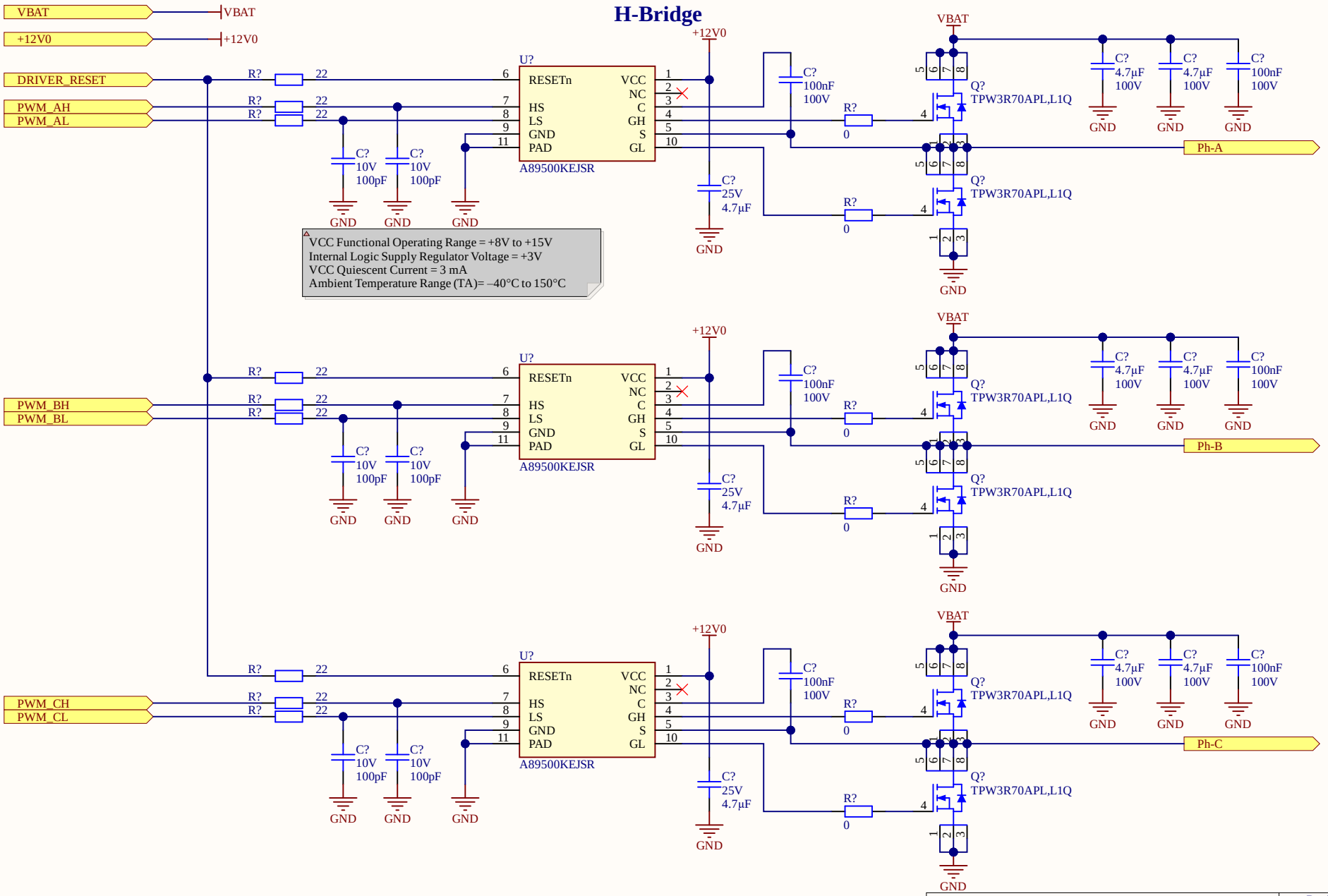


Title	***MOTORCONTROLLER***		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:45	Sheet 7 of 16	

ACCELEROMETERS



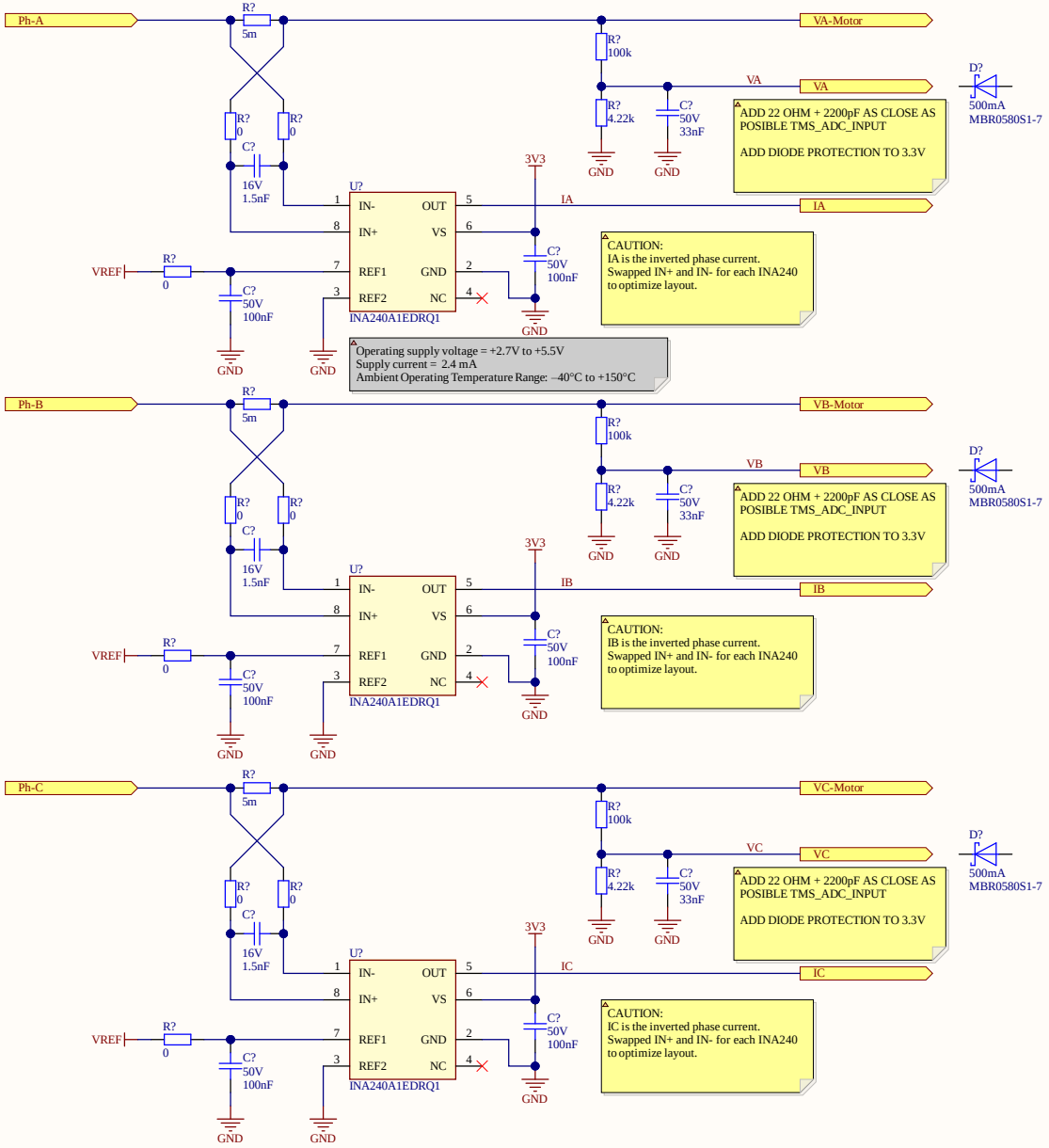
Title	**MOTORCONTROLLER**		Droid Technologies
Size: A4	Number: DT10965	Revision: 1.0	
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H-Bridge
VBAT
+12V0
DRIVER_RESET
PWM_AH
PWM_AL
PWM_BH
PWM_BL
PWM_CH
PWM_CL
U?
RESETn
HS
LS
GND
PAD
VCC
NC
C
GH
S
GL
A89500KEJSR
R?
22
C?
10V
100pF
C?
25V
4.7µF
C?
100nF
100V
C?
4.7µF
100V
R?
0
Q?
TPW3R70APL,L1Q
Ph-A
Ph-B
Ph-C
VCC Functional Operating Range = +8V to +15V
Internal Logic Supply Regulator Voltage = +3V
VCC Quiescent Current = 3 mA
Ambient Temperature Range (TA)= –40°C to 150°C
Title MOTORCONTROLLER
Size: A4
Number: DT10965
Revision: 1.0
Date: 16.06.2025
Time: 12:45:45
Sheet 9 of 16
Droid Technologies

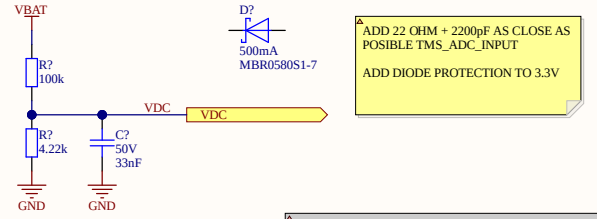
Phase Current/Voltage Sense

3V3 3V3
5V0 5V0
VBAT VBAT

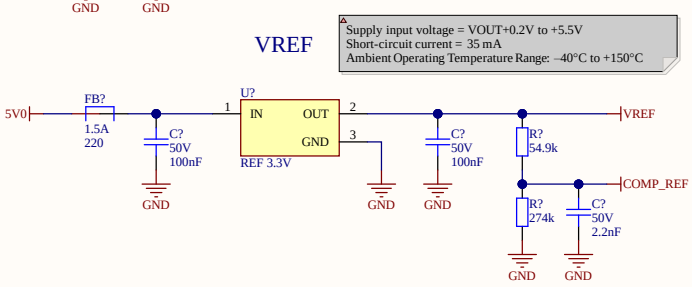
Phase Current/Voltage Sense



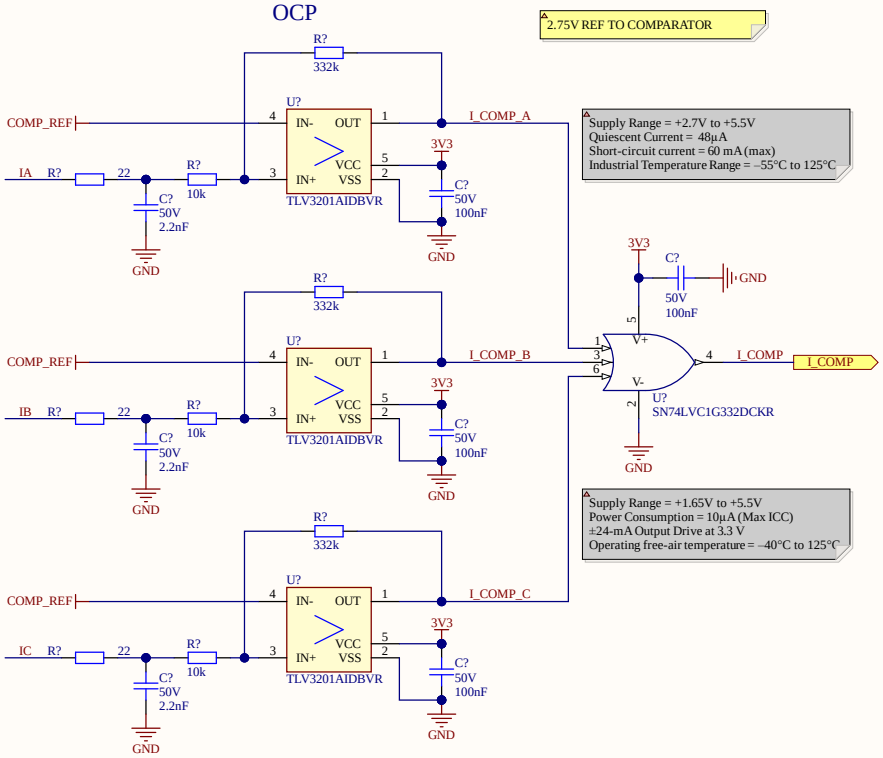
VBUS Sense



VREF

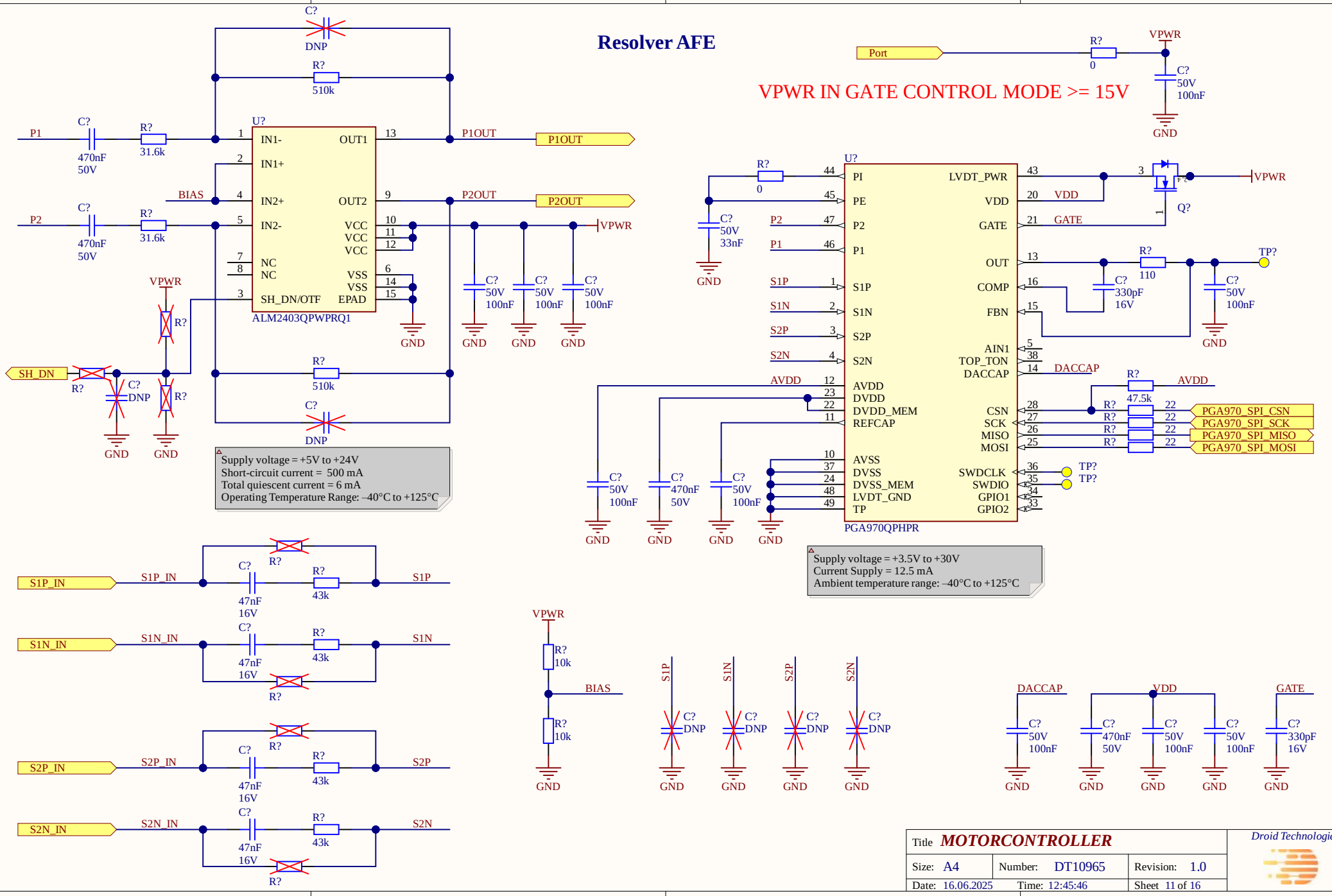


OCP



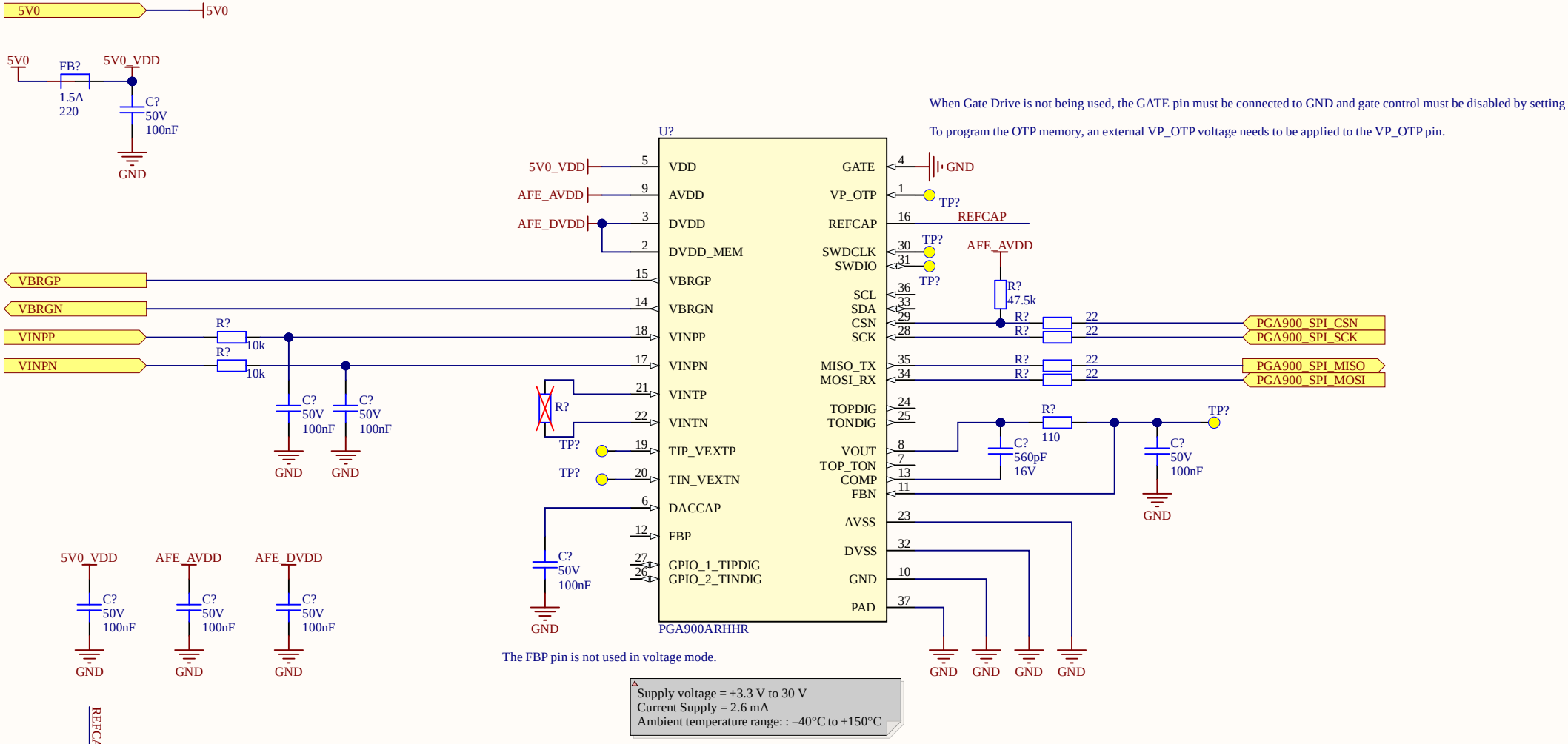
Title	MOTORCONTROLLER		Droid Technologies
Size: A3	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:45	Sheet 10 of 16	

Resolver AFE



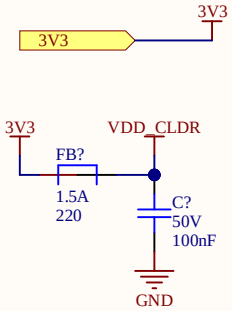
Title	**MOTORCONTROLLER**		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:46	Sheet 11 of 16	

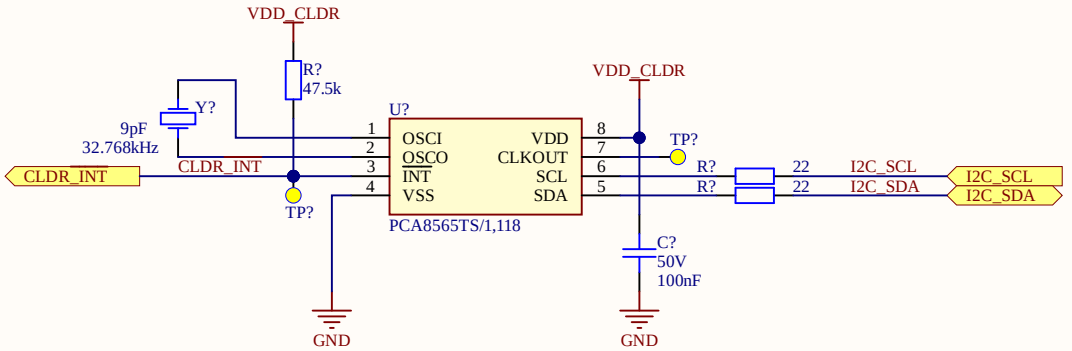
Pressure sensor AFE



Title	**MOTORCONTROLLER**		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:46	Sheet 12 of 16	

CALENDAR

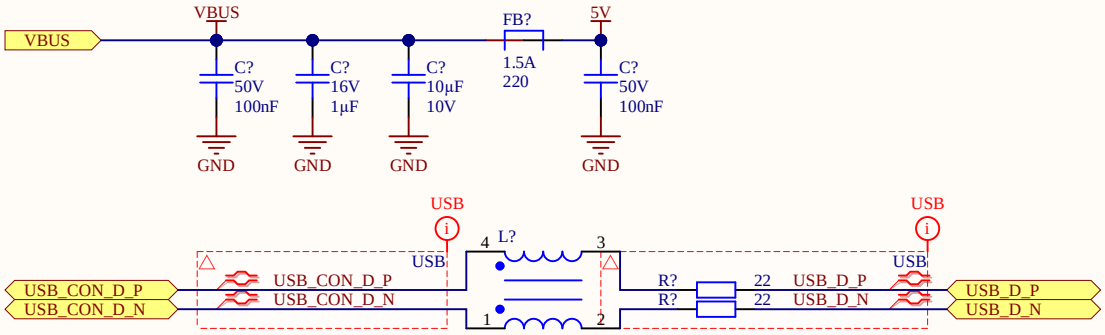


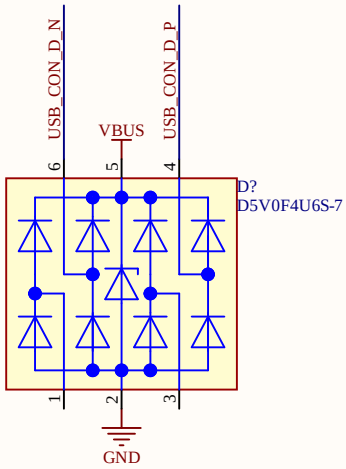


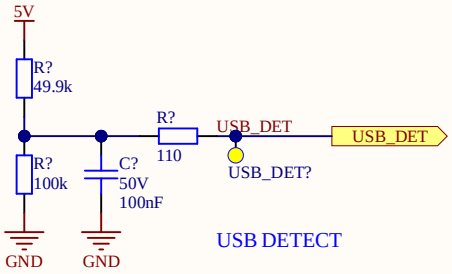
Supply voltage = : 0.9 V to 5.5 V
Current Supply = 820 µA
Ambient temperature range: : –40°C to +150°C

Title	***MOTORCONTROLLER***		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
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USB INTERFACE



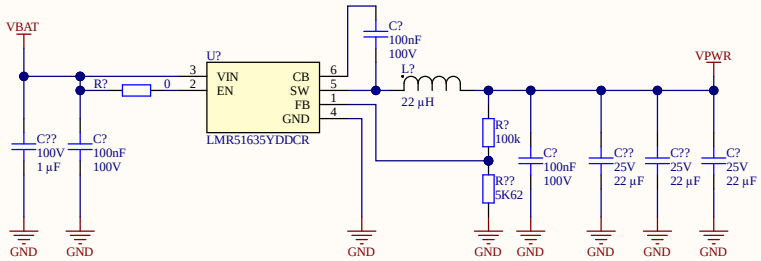




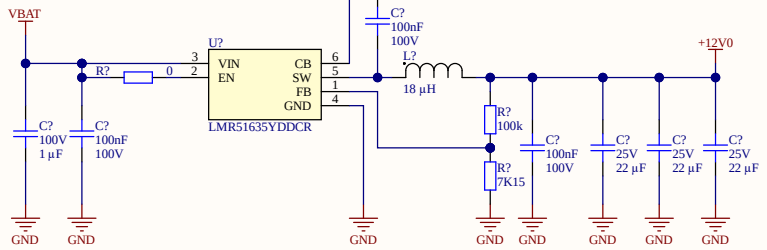
Title **MOTORCONTROLLER**			*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:46	Sheet 14 of 16	

VBAT
+12V0
VPWR
5V0
3V3
VDD_1V2

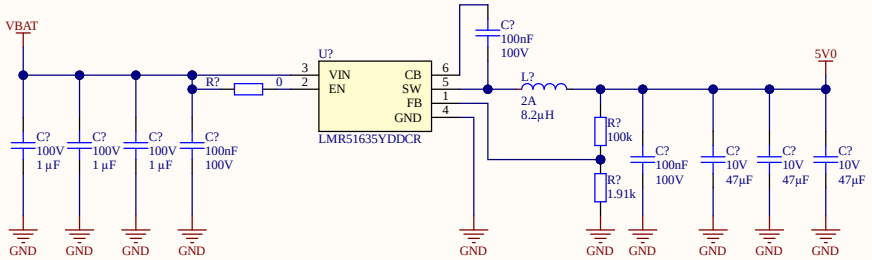
VPWR (for resolver)
VBAT(up to +60V) --> VPWR (+15V)



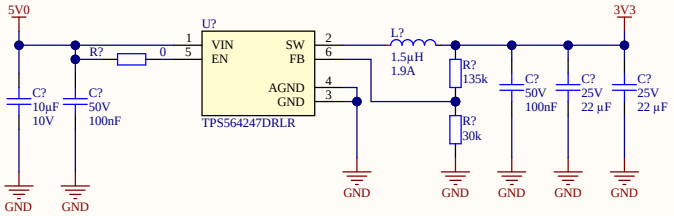
+12V for Motor control (H-Bridge)
VBAT(up to +60V) --> +12V



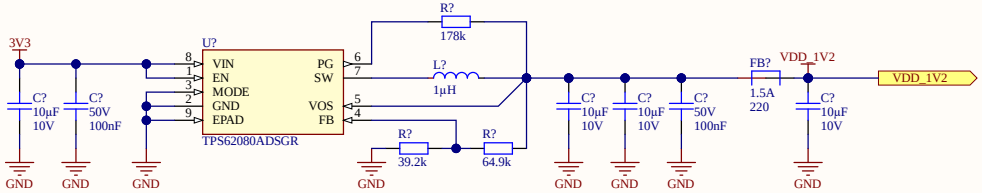
VBAT(up to +60V) --> +5V0



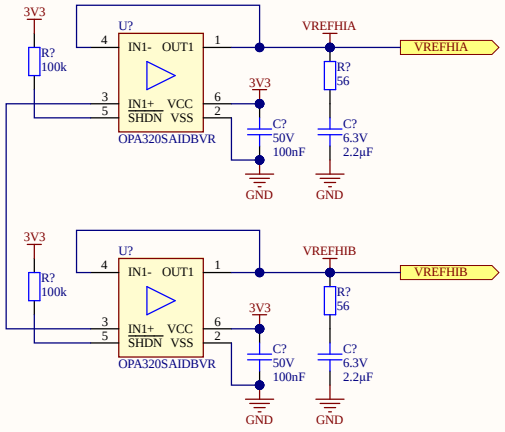
+5V0 --> +3V3



+3V3 --> +1V2 (Vdd for MCU)



Supply voltage = +1.8 V to +5.5 V
Current Supply = 1.45 mA
Ambient temperature range: : –40°C to +150°C



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Battery

48V 14S LiFePO4 battery

(about 51.38V@3.68V per cell)

MOTOR

Maxon EC-4pole 32

brushless DC 32mm motor
Part number 397800

Resolver

*

*

*

Pressure Sensor

KELLER Series 7L/LHP

*

*

Title	***MOTORCONTROLLER***		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 16.06.2025	Time: 12:45:46	Sheet 16 of 16	